

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1558

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I M.Mary Sharlin (192511330) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: M.Mary Sharlin

Assessment Tasks

Industry Problem Solving Task

- 1. Retail Data Optimization Problem - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.**

ANSWER:

Problem Analysis

The retail chain receives data from physical stores, online sales, inventory systems, customer applications, loyalty programs, and websites. The main challenges are integrating these different data sources, tracking inventory in real time, forecasting demand, and providing personalized recommendations.

Proposed Big Data Architecture

Data Sources → Data Ingestion → Data Lake → Processing → Analytics/ML → Applications

1. Data Sources

- Point-of-sale (POS) transactions
- E-commerce transactions
- Inventory and warehouse systems

- Customer browsing history
- Loyalty-card data
- Product reviews
- Supplier information

2. Data Ingestion

Use Apache Kafka for continuous streaming of sales and inventory events. Batch data can be loaded through ETL/ELT pipelines.

3. Data Storage

Use a cloud data lake or Hadoop-based storage to store:

- Structured sales data
- Semi-structured customer data
- Unstructured reviews and logs

A data warehouse can be used for highly structured reporting and business intelligence.

4. Data Processing

Apache Spark can perform large-scale processing, while Spark Structured Streaming can process inventory and sales events in near real time.

For example:

Sale occurs → Kafka → Stream Processing → Inventory database → Inventory dashboard

The inventory level can therefore be updated almost immediately.

5. Analytics Models

Demand Forecasting

- ARIMA
- Prophet
- Random Forest
- Gradient Boosting
- LSTM for complex time-series patterns

Factors such as previous sales, seasonality, holidays, promotions, and location can be used.

Personalized Recommendations

- Collaborative filtering
- Content-based filtering
- Hybrid recommendation systems

6. Applications

The final system provides:

- Real-time inventory tracking
- Automatic low-stock alerts
- Demand forecasts
- Personalized product recommendations
- Customer segmentation
- Management dashboards

Conclusion

A combination of Kafka + data lake + Spark + machine learning + recommendation models provides a scalable architecture. It helps prevent stockouts, reduces excess inventory, improves demand planning, and increases customer engagement through personalized marketing.

2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

ANSWER:

Problem Analysis

A social media platform generates billions of events including posts, comments, likes, follows, shares, and login activities. Processing all these events using traditional batch systems would cause significant delays.

The system therefore requires distributed storage, stream processing, machine learning, and scalable infrastructure.

Proposed Architecture

User Activity → Kafka → Stream Processing → Feature Extraction → ML Models → Risk Score → Action

1. Data Sources

- Posts and comments
- Likes and shares
- Account creation information
- Login locations
- Device information
- Follower/following relationships
- Posting frequency
- URLs and hashtags

2. Stream Processing

Apache Kafka can act as the event-streaming platform.

Apache Flink or Spark Structured Streaming can process incoming events continuously.

For example:

Account creates 100 posts in one minute → Stream processor detects abnormal activity → ML model calculates risk → Account is flagged

3. Fake Account Detection

Machine-learning features can include:

- Account age
- Number of followers
- Followers/following ratio
- Posting frequency
- Login patterns
- Device similarity
- Network behavior

Models such as:

- Random Forest
- Gradient Boosting
- Neural Networks

can classify accounts as normal or suspicious.

4. Misinformation Detection

Natural Language Processing (NLP) can analyze:

- Text content
- Sentiment
- Keywords
- Repeated claims
- Source credibility
- URL patterns

A classification model can assign a probability that content violates platform misinformation policies. Graph analytics can also identify coordinated networks of accounts sharing the same content.

5. Scalability

Distributed systems divide the workload among multiple machines:

Billions of Events → Partitioning → Multiple Processing Nodes → Parallel Analysis

This allows the platform to process very large workloads without depending on a single server.

Conclusion

Combining Kafka, Flink/Spark, distributed storage, machine learning, NLP, and graph analytics enables rapid detection of suspicious accounts and coordinated harmful activity. Human review can be incorporated for high-impact decisions to reduce false positives.

3. Government Data Platform Problem A government agency is building a national data platform.

It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

ANSWER:

Problem Analysis

A government national data platform may combine:

- Census information
- Economic statistics
- Geographic data
- Healthcare information
- Education datasets
- Infrastructure information

Different departments may use different formats, identifiers, databases, and standards. The major challenge is achieving interoperability while protecting sensitive information.

Proposed Architecture

Departmental Systems → Data Integration Layer → National Data Lake/Warehouse → Governance Layer → Authorized Applications

1. Data Integration

Use APIs, ETL/ELT pipelines, and event-based integration to collect information from different departments.

A common data model should define:

- Data formats
- Naming conventions
- Identifiers
- Metadata
- Data relationships

2. Master Data Management

Common entities such as:

- Citizen
- Organization
- Location
- Government department

should have standardized identifiers.

This prevents the same entity from being represented differently across departments.

3. Data Storage

A government data lake can store raw and semi-structured data, while curated datasets can be maintained in a data warehouse.

Geospatial databases can be used for geographic information.

4. Data Governance

A centralized governance framework should define:

- Data ownership
- Data stewardship
- Data quality standards
- Metadata standards
- Data classification
- Data retention
- Data-sharing policies

5. Privacy and Security

Sensitive information should be protected using:

- Encryption
- Role-based access control
- Multi-factor authentication
- Data masking
- Anonymization/pseudonymization
- Audit logs
- Network security
- Continuous monitoring

Access should follow the least-privilege principle.

6. Interoperability

Common APIs and standardized formats allow different departments to exchange information.

For example:

Department A → Standard API → Integration Layer → Department B

This eliminates isolated data silos.

Conclusion

A combination of common data standards, APIs, master data management, metadata governance, secure data sharing, and strong access controls creates an integrated national data platform while maintaining privacy and security.

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

ANSWER:

Problem Analysis

A bank processes millions of transactions every minute. Fraud detection must therefore operate with very low latency, while considering both historical customer behavior and current transaction activity.

Proposed Big Data Pipeline

Transactions → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud Score → Decision
Historical data is stored separately and used to train machine-learning models.

1. Data Sources

The system analyzes:

- Transaction amount
- Account information
- Merchant
- Location
- Device
- IP address
- Time
- Previous transactions
- Login history

2. Data Ingestion

Apache Kafka can ingest millions of transaction events and distribute them across processing partitions.

3. Stream Processing

Apache Flink or Spark Structured Streaming can calculate real-time features such as:

- Number of transactions in the last 5 minutes
- Average transaction amount
- Sudden location changes
- Unusual merchant activity
- Multiple failed transactions
- New device usage

4. Machine-Learning Models

Possible models include:

Random Forest / Gradient Boosting

Useful for supervised fraud classification.

Logistic Regression

Provides a relatively interpretable baseline.

Autoencoders or other anomaly-detection methods

Useful for detecting unusual behavior when labeled fraud examples are limited.

5. Decision Engine

The model produces a fraud probability.

Example:

$\text{Risk} < \text{Threshold 1} \rightarrow \text{Approve}$

Threshold 1–Threshold 2 → Additional authentication

Risk > Threshold 2 → Block/Investigate

6. Historical + Streaming Data

Historical data establishes the customer's normal behavior.

Streaming data determines whether the current transaction deviates from that behavior.

For example, if a customer normally spends ₹2,000 locally but suddenly makes several high-value transactions from different locations within minutes, the system can identify the behavior as suspicious.

Conclusion

The combination of Kafka, Flink/Spark, historical data, real-time feature engineering, anomaly detection, and supervised machine learning provides fast and accurate fraud detection. It minimizes transaction delays while reducing financial losses.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

ANSWER:

Problem Analysis

Smart farming generates data from soil sensors, weather stations, drones, satellites, agricultural equipment, and crop-monitoring systems. These sources are distributed and generate data continuously.

The objective is to improve:

- Crop yield
- Water usage
- Fertilizer usage
- Pest management
- Resource efficiency

Proposed Big Data Architecture

Sensors/Drones/Satellites → IoT Gateway → Kafka → Data Lake → Spark/ML → Decision Support System

1. Data Sources

Soil Sensors

- Soil moisture
- Temperature
- pH
- Nutrient levels

Weather Sensors

- Temperature
- Humidity
- Rainfall
- Wind speed

Drones/Satellites

- Crop images
- Vegetation indices
- Plant health information

Farm Equipment

- Fuel consumption
- Machine location
- Operating conditions

2. Data Ingestion

IoT gateways collect sensor information.

Kafka can stream continuous sensor data into the big data platform.

3. Data Storage

A distributed data lake can store:

- Sensor readings
- Historical crop data
- Weather data
- Drone images
- Satellite imagery

4. Data Processing

Apache Spark can process large datasets and identify relationships between environmental conditions and crop performance.

5. Analytics and Machine Learning

Yield Prediction

Machine-learning models can predict expected crop yield using:

- Soil conditions
- Weather
- Historical yield
- Crop type
- Fertilizer usage

Possible models include:

- Random Forest
- Gradient Boosting
- Neural Networks

Irrigation Optimization

If soil moisture falls below an appropriate threshold and weather data indicates low rainfall probability, the system can recommend irrigation.

Disease Detection

Drone images can be analyzed using computer-vision and deep-learning models to identify abnormal crop patterns.

6. Decision Support

The system can provide recommendations such as:

- Irrigate Field A
- Reduce fertilizer in Field B
- Inspect Field C for possible disease
- Expected yield: X tonnes/hectare

Benefits

Big data analytics enables:

- Higher crop yield
- Reduced water consumption
- Efficient fertilizer usage
- Early disease detection
- Reduced operating costs
- Better resource allocation

Conclusion

A combination of IoT sensors, Kafka, distributed data lakes, Spark, machine learning, and drone/satellite analytics creates a scalable precision-agriculture system. Real-time analytics allows farmers to make faster, data-driven decisions and use agricultural resources more efficiently.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded: